Mathlearn · 8-sinf · Algebra

Sonning moduli. Modul qatnashgan tenglama va tengsizliklar

1-test (oson) · 20 ta savol

Ism-familiya: _____ Sinf: _____ Sana: _____

Har savolda bitta to'g'ri javobni belgilang.

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1. $|-13|$ **nimaga teng?**

A) 0 **B)** 13 **C)** $\frac{1}{13}$ **D)** -13

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2. **Qaysi tenglik to'g'ri?**

A) $-|5| = 5$ **B)** $|-5| = -5$

C) $|-5| = 5$ **D)** $|5| = -5$

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3. $a < 0$ **bo'lsa,** $|a|$ **nimaga teng?**

A) $-a$ **B)** $a - 1$ **C)** a **D)** 0

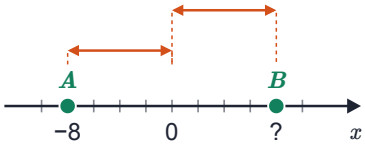
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4. $|-2{,}4| + |1{,}6|$ **ni hisoblang.**

A) -4 **B)** $0{,}8$ **C)** $-0{,}8$ **D)** 4

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5. A **va** B **nuqtalar 0 dan bir xil masofada yotibdi.** B **nuqtaning koordinatasi qanday?**



A) 8 **B)** 0

C) 16 **D)** -8

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6. **Rasmda qaysi tengsizlikning yechimlari tasvirlangan?**



A) $|x| < 3$ **B)** $|x| \geq 3$

C) $|x| \leq 3$ **D)** $|x| > 3$

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7. $|x| = 11$ **tenglamaning ildizlarini toping.**

A) 11 va -11 **B)** faqat -11

C) ildiz yo'q **D)** faqat 11

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8. $|x| = -4$ **tenglamaning ildizlarini toping.**

A) faqat -4 **B)** ildiz yo'q

C) faqat 4 **D)** 4 va -4

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9. $|x - 6| = 0$ **tenglamani yeching.**

A) -6 **B)** 6 **C)** 6 va -6 **D)** 0

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10. $|x + 2| = 5$ **tenglamani yeching.**

A) -3 va 7 **B)** 3 va 7 **C)** 3 va -7 **D)** faqat 3

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11. $|x| < 5$ **tengsizlik qaysi qo'sh tengsizlikni bildiradi?**

A) $-5 < x < 5$ **B)** $0 < x < 5$ **C)** $-5 \le x \le 5$ **D)** $x < 5$

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12. $-2{,}7 \le x \le 2{,}7$ **qo'sh tengsizlikni modul bilan yozing.**

A) $|x| \le 2{,}7$ **B)** $|x| < 2{,}7$ **C)** $|x| \ge 2{,}7$ **D)** $|x| = 2{,}7$

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13. $|x| \ge 6$ **tengsizlikning yechimlarini toping.**

A) $-6 \le x \le 6$

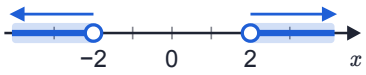
B) $x \le -6$ yoki $x \ge 6$

C) $x < -6$ yoki $x > 6$

D) faqat $x \ge 6$

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14. Rasmda qaysi tengsizlikning yechimlari tasvirlangan?



A) $|x| > 2$ **B)** $|x| < 2$

C) $|x| \ge 2$ **D)** $|x| \le 2$

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15. $|x| \le 0$ **tengsizlikning yechimlari qanday?**

A) istalgan son

B) faqat $x = 0$

C) yechim yo'q

D) $x \ge 0$

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16. $|x| \ge -3$ **tengsizlikning yechimlari qanday?**

A) istalgan son

B) $x \ge 3$

C) $x \le -3$ yoki $x \ge 3$

D) yechim yo'q

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17. $|x| < 3{,}5$ **tengsizlikning nechta butun yechimi bor?**

A) 8 **B)** 7 **C)** 6 **D)** 3

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18. Qaysi son $|x| \le 4$ **tengsizlikning yechimi bo'ladi?**

A) 7 **B)** 5 **C)** $-4{,}1$ **D)** -4

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19. $|x - 3| \le 1$ **tengsizlikni yeching.**

A) $x \le 4$

B) $-2 \le x \le 4$

C) $x \le 2$ yoki $x \ge 4$

D) $2 \le x \le 4$

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20. $|x - 5|$ **son o'qida nimani bildiradi?**

A) x ni 5 marta kattalashtirishni

B) x va 5 nuqtalar orasidagi masofani

C) x va -5 nuqtalar orasidagi masofani

D) 0 dan x gacha masofani

Javoblar

Sonning moduli. Modul qatnashgan tenglama va tengsizliklar · 1-test (oson)

№	Javob	Qiymati
1	**B**	13
2	**C**	$\lvert -5\rvert = 5$
3	**A**	$-a$
4	**D**	4
5	**A**	8
6	**C**	$\lvert x\rvert \le 3$
7	**A**	11 va -11
8	**B**	ildiz yo'q
9	**B**	6
10	**C**	3 va -7

№	Javob	Qiymati
11	**A**	$-5 < x < 5$
12	**A**	$\lvert x\rvert \le 2{,}7$
13	**B**	$x \le -6$ yoki $x \ge 6$
14	**A**	$\lvert x\rvert > 2$
15	**B**	faqat $x = 0$
16	**A**	istalgan son
17	**B**	7
18	**D**	-4
19	**D**	$2 \le x \le 4$
20	**B**	x va 5 nuqtalar orasidagi masofani

Qisqa yechimlar

1. Manfiy sonning moduli unga qarama-qarshi songa teng: -13 ning qarama-qarshisi **13**.

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2. Modul manfiy bo'lmaydi: $\lvert -5\rvert = 5$ va $\lvert 5\rvert = 5$.
$-\lvert 5\rvert = -5$, 5 emas. To'g'risi: $\lvert -5\rvert = 5$.

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3. Ta'rif: $a < 0$ bo'lsa, $\lvert a\rvert = -a$. Masalan, $a = -4$ da $\lvert -4\rvert = 4 = -(-4)$. Javob: $-a$.

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4. $\lvert -2{,}4\rvert = 2{,}4$, $\lvert 1{,}6\rvert = 1{,}6$.
$2{,}4 + 1{,}6 = $ **4**.

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5. A nuqta 0 dan $\lvert -8\rvert = 8$ birlik chapda.
B esa 0 dan 8 birlik o'ngda: **8**.

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6. Bo'yalgan kesma -3 dan 3 gacha — 0 dan 3 birlikdan uzoq bo'lmagan nuqtalar.
Uchlari bo'yalgan (kiradi), demak belgi «$\le$»: $\lvert x\rvert \le 3$.

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7. 0 dan 11 birlik uzoqlikda ikkita son bor: 11 va -11.
Javob: **11 va** -11.

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8. Modul manfiy son bo'lmaydi, u -4 ga teng bo'la olmaydi.
Javob: **ildiz yo'q**.

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9. Modul faqat ichidagi ifoda 0 bo'lganda 0 ga teng: $x - 6 = 0$.
$x = $ **6**.

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10. $x + 2 = 5$, bundan $x = 3$.
$x + 2 = -5$, bundan $x = -7$.
Javob: **3 va** -7.

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11. $\lvert x\rvert < a$ tengsizlik $-a < x < a$ ni bildiradi.
Belgi qat'iy, uchlari kirmaydi: $-5 < x < 5$.

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12. Chetlari qarama-qarshi sonlar, belgilar «$\le$».
Demak: $\lvert x\rvert \le 2{,}7$.

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13. 0 dan kamida 6 birlik uzoqdagi nuqtalar ikki nurda yotadi.
Belgi «$\geq$», ± 6 ham kiradi: $x \leq -6$ **yoki** $x \geq 6$.

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14. Ikkita nur — 0 dan uzoqdagi nuqtalar, belgi «$>$» yoki «$\geq$».
-2 va 2 bo'sh doiracha — kirmaydi: $|x| > 2$.

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15. Modul manfiy bo'lmaydi, demak $|x| = 0$ bo'lishi kerak.
Yagona yechim: $x = 0$.

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16. Modul doim ≥ 0, demak har doim -3 dan katta.
Javob: **istalgan son**.

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17. $-3{,}5 < x < 3{,}5$.
Butun sonlar: $-3;\ -2;\ -1;\ 0;\ 1;\ 2;\ 3$ — **7 ta**.

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18. $|-4| = 4$, $4 \leq 4$ — to'g'ri.
$|5| = 5$, $|-4{,}1| = 4{,}1$, $|7| = 7$ — hammasi 4 dan katta. Javob: -4.

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19. $-1 \leq x - 3 \leq 1$.
Hamma qismga 3 ni qo'shamiz: $2 \leq x \leq 4$.

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20. $|x - a|$ — x va a nuqtalar orasidagi masofa.
Bu yerda $a = 5$: x **va 5 orasidagi masofa**.
